

# What Does Scalar RLHF Optimize?

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- ▶ Uniform logistic reward fitting implements the population Borda ordering.
- ▶ KL-regularized policy optimization exponentially tilts a reference policy toward the fitted rewards.
- ▶ Changing how comparison labels are processed can change the social-choice rule.

# RLHF Does Not Observe Harsanyi's Cardinal Utilities

The previous lecture began with preferences over lotteries satisfying the VnM axioms and derived cardinal preferences represented on different interpersonal scales:

$$u_1, \dots, u_n.$$

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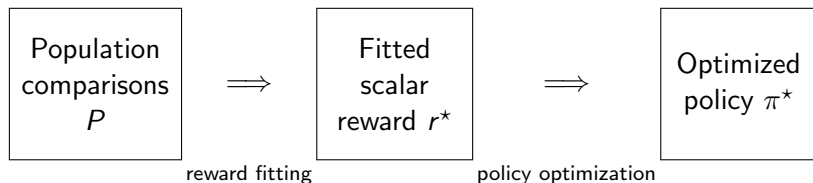
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RLHF nevertheless constructs and optimizes a scalar reward.  
Which aggregate objective has it constructed?



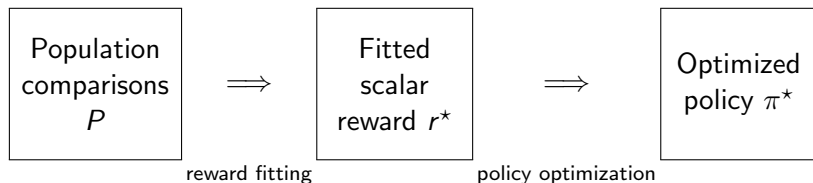
# We Will Analyze Two Maps in the RLHF Pipeline

Fix one prompt and a finite set  $C$  of candidate completions.



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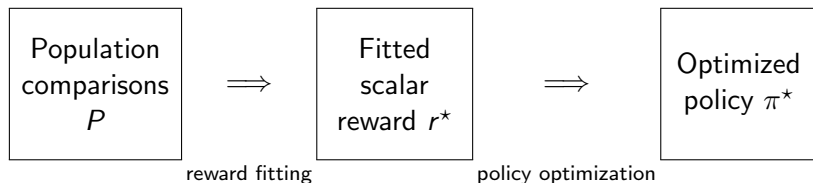
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# We Will Analyze Two Maps in the RLHF Pipeline

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The first map depends on the loss, pair-sampling design, label processing, and reward class.

The second map depends on the fitted reward, reference policy, and regularization strength.

# A Clean Population Model Isolates the Aggregation Rule

We make four simplifying assumptions:

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2. Every unordered pair is presented for labeling with equal probability.
3. We work with the population comparison probabilities, not a finite sample.
4. The reward vector  $r \in \mathbb{R}^m$  has one free score  $r_i$  for each response.

These assumptions remove statistical and approximation questions so that we can see the social aggregation performed by the objective itself.

# Population Comparisons Form a Probabilistic Tournament

For distinct responses  $i, j \in C$ , define

$$P_{ij} := \Pr(i \text{ is preferred to } j \mid \{i, j\} \text{ is presented}).$$

Because one response wins each comparison,  $P_{ij} + P_{ji} = 1$ .



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Because one response wins each comparison,  $P_{ij} + P_{ji} = 1$ . The randomness can come from heterogeneous labelers or from variation in repeated judgments.

## Recall the Cyclic Majority Example

Three equally common preference types rank three responses as

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No single scalar ordering can reproduce all three comparisons.

# A Scalar Reward Predicts a Transitive Tournament

Let

$$\sigma(z) := \frac{1}{1 + e^{-z}}$$

denote the logistic (sigmoid) function. A Bradley–Terry reward vector  $r \in \mathbb{R}^m$  predicts

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The population matrix  $P$  can cycle, but the fitted matrix  $Q^r$  must inherit one transitive ordering from  $r$ .

The fitting loss decides how the cycle is compressed.



# A Comparison Label Has a Bernoulli Likelihood

Fix two responses  $i$  and  $j$ . Let  $Y = 1$  when  $i$  is preferred and  $Y = 0$  when  $j$  is preferred, and write

$$p := \Pr(Y = 1), \quad z := r_i - r_j.$$

Thus  $Y \sim \text{Bernoulli}(p)$  in the population.

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The Bradley–Terry model assigns

$$\Pr_z(Y = 1) = \sigma(z), \quad \Pr_z(Y = 0) = \sigma(-z).$$

Hence the likelihood of an observed label  $y \in \{0, 1\}$  is

$$\text{Lik}(z; y) := \Pr_z(Y = y) = \sigma(z)^y \sigma(-z)^{1-y}.$$

# Expected Negative Log-Likelihood Is Logistic Loss

Taking logs gives

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Taking the negative expectation under the population label  $Y \sim \text{Bernoulli}(p)$  gives

$$\begin{aligned}\ell(p, z) &:= -\mathbb{E}[\log \text{Lik}(z; Y)] \\ &= -p \log \sigma(z) - (1 - p) \log \sigma(-z) \\ &= p \log(1 + e^{-z}) + (1 - p) \log(1 + e^z).\end{aligned}$$

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Since

$$\log(1 + e^{-z}) = \log(1 + e^z) - z,$$

we obtain the useful form

$$\ell(p, z) = \log(1 + e^z) - pz.$$

# Uniform Pair Sampling Gives One Population Objective

When every unordered pair receives equal weight, reward fitting minimizes

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Only reward differences matter:

$$L(r + c\mathbf{1}) = L(r).$$

Thus the fitted scores are identified only up to an additive constant. This does not affect their ordering or any predicted comparison probability.



# Borda Measures Average Pairwise Success

## Definition

The *population Borda total* of response  $i$  is

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Its normalized Borda score is

$$B_i := \frac{S_i}{m-1}.$$

Thus  $B_i$  is the probability that  $i$  defeats an opponent drawn uniformly from the other responses. The totals  $S_i$  and scores  $B_i$  induce the same ordering.

# Uniform Logistic Fitting Implements Borda

## Theorem

*Let  $r^*$  be any minimizer of the uniformly weighted logistic loss  $L$ .  
Then, for every  $i, k$ ,*

$$r_i^* > r_k^* \iff S_i > S_k.$$

*The same equivalence holds with  $>$  replaced by  $=$ .*

# Proof Roadmap: Optimality Becomes Moment Matching

1. Differentiate the logistic loss:  $\partial_z \ell(p, z) = \sigma(z) - p$ , the model win probability minus the population win probability.

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1. Differentiate the logistic loss:  $\partial_z \ell(p, z) = \sigma(z) - p$ , the model win probability minus the population win probability.
2. Use the first-order conditions to match each response's population and predicted average win rate.
3. Use monotonicity of  $\sigma$ : a higher fitted score gives a higher predicted win probability against every fixed comparison score, and hence a larger average win rate.

# The Derivative Measures the Probability Gap

Recall that  $\ell(p, z) = \log(1 + e^z) - pz$ . Treating  $p$  as fixed,

$$\begin{aligned}\frac{\partial}{\partial z} \ell(p, z) &= \frac{\partial}{\partial z} [\log(1 + e^z) - pz] \\ &= \frac{e^z}{1 + e^z} - p \\ &= \frac{1}{1 + e^{-z}} - p \\ &= \sigma(z) - p.\end{aligned}$$



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Here  $\sigma(z)$  is the model probability of  $Y = 1$ , while  $p$  is its population probability.

# Each Pair Contributes a Probability Gap

For a pair written as  $i < j$ , its loss is  $\ell(P_{ij}, r_i - r_j)$ . The chain rule gives

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$$\begin{aligned}\frac{\partial}{\partial r_i} \ell(P_{ij}, r_i - r_j) &= \left. \frac{\partial \ell(p, z)}{\partial z} \right|_{\substack{p=P_{ij} \\ z=r_i-r_j}} \frac{\partial (r_i - r_j)}{\partial r_i} \\ &= (\sigma(r_i - r_j) - P_{ij}) \cdot 1 \\ &= \sigma(r_i - r_j) - P_{ij}.\end{aligned}$$

# The Pair Formula Does Not Depend on Its Written Order

If  $j < i$ , the pair appears as  $\ell(P_{ji}, r_j - r_i)$ . Its contribution is

$$\begin{aligned}\frac{\partial}{\partial r_i} \ell(P_{ji}, r_j - r_i) &= (\sigma(r_j - r_i) - P_{ji}) \frac{\partial(r_j - r_i)}{\partial r_i} \\ &= (\sigma(r_j - r_i) - P_{ji})(-1) \\ &= -((1 - \sigma(r_i - r_j)) - (1 - P_{ij})) \\ &= \sigma(r_i - r_j) - P_{ij}.\end{aligned}$$

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Thus, for a fixed coordinate  $r_i$ , every opponent contributes the same formula regardless of how the unordered pair was written in  $L$ :

$$\frac{\partial L}{\partial r_i}(r) = \sum_{j \neq i} (\sigma(r_i - r_j) - P_{ij}).$$

## Optimal Fitting Matches Total Wins

Because  $r^*$  is an unconstrained minimizer of the differentiable function  $L$ , its gradient vanishes. The  $i$ th first-order condition is therefore

$$\begin{aligned} 0 &= \frac{\partial L}{\partial r_i}(r^*) \\ &= \sum_{j \neq i} (Q_{ij}^{r^*} - P_{ij}). \end{aligned}$$

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Set

$$P_{ii} = Q_{ii}^r = \frac{1}{2}.$$

Adding the same diagonal term to both sides gives

$$\sum_{j \in C} Q_{ij}^{r^*} = \sum_{j \in C} P_{ij} = S_i + \frac{1}{2}.$$

# Bradley–Terry Row Totals Reveal the Scores

Suppose  $r_i > r_k$ . For every fixed opponent  $j$ ,

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Equal scores give equal row totals, while reversing the score inequality reverses every termwise comparison. Therefore

$$r_i > r_k \iff \sum_j Q_{ij}^r > \sum_j Q_{kj}^r.$$

# Moment Matching Proves the Borda Ordering

Combine the preceding two conclusions:

$$\begin{aligned}r_i^* > r_k^* &\iff \sum_j Q_{ij}^{r^*} > \sum_j Q_{kj}^{r^*} \\&\iff S_i + \frac{1}{2} > S_k + \frac{1}{2} \\&\iff S_i > S_k.\end{aligned}$$

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So we learned rewards embed the Borda ranking.

# The Running Cycle Is Compressed to a Tie

For the equal mixture of the three cyclic rankings,

$$S_a = P_{ab} + P_{ac} = \frac{2}{3} + \frac{1}{3} = 1.$$

The same calculation holds cyclically for  $b$  and  $c$ , so  $S_a = S_b = S_c = 1$ . The Borda theorem therefore gives

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$$r_a^* = r_b^* = r_c^*.$$

All fitted score differences are therefore zero, so the fitted model predicts  $Q_{ij}^{r^*} = 1/2$  on every pair.

# The Theorem Gives an Ordering, Not More

The Borda theorem identifies its ordering, but generally

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The numerical gaps are determined jointly by all pairwise comparisons.

Nor has the fit recovered Harsanyi welfare or a representative person's utility. Pairwise comparison data did not contain interpersonal utility scales in the first place.

## The Second Stage Balances Reward and Policy Change

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RLHF instead begins from a reference policy  $\pi_0$ , representing the model before reward optimization, and trades reward against changing that policy. We therefore need a numerical measure of departure from  $\pi_0$ .

# KL Penalizes Departure from the Reference Policy

For a policy  $\pi \in \Delta(C)$  and a full-support reference policy  $\pi_0 \in \Delta(C)$ , define

$$\text{KL}(\pi \parallel \pi_0) := \sum_i \pi(i) \log \frac{\pi(i)}{\pi_0(i)}.$$

Terms with  $\pi(i) = 0$  are defined to be zero. The full-support assumption on  $\pi_0$  keeps every denominator positive.

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KL divergence is nonnegative and equals zero only when  $\pi = \pi_0$ .

# Policy Optimization Trades Reward Against Policy Change

Let  $\pi_0 \in \Delta(C)$  be a full-support reference policy, and let  $\tau > 0$ .  
The policy-optimization stage solves

$$\max_{\pi \in \Delta(C)} \left\{ \sum_i \pi(i) r_i - \tau \text{KL}(\pi \| \pi_0) \right\}.$$

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The first term rewards high-scoring responses.

The second penalizes movement away from  $\pi_0$ . The temperature  $\tau$  controls the strength of this penalty.

# The Optimized Policy Is an Exponential Tilt

## Theorem

*The unique optimizer of the KL-regularized policy objective is*

$$\pi_r(i) = \frac{\pi_0(i)e^{r_i/\tau}}{\sum_j \pi_0(j)e^{r_j/\tau}}.$$

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We first prove the formula, then use it to connect the learned social order to the optimized policy's behavior.

# Proof: Rewrite the Candidate's Log Ratio

Define

$$Z_r := \sum_j \pi_0(j) e^{r_j/\tau}, \quad \pi_r(i) := \frac{\pi_0(i) e^{r_i/\tau}}{Z_r}.$$

Then

$$\log \frac{\pi_r(i)}{\pi_0(i)} = \frac{r_i}{\tau} - \log Z_r.$$

## Proof: KL Nonnegativity Gives Optimality

For each response  $i$ , inserting  $\pi_0(i)$  gives

$\pi(i)/\pi_r(i) = [\pi(i)/\pi_0(i)]/[\pi_r(i)/\pi_0(i)]$ . The denominator ratio is divided, so logarithms subtract. Therefore

$$\begin{aligned}\text{KL}(\pi\|\pi_r) &= \sum_i \pi(i) \log \frac{\pi(i)}{\pi_r(i)} \\ &= \sum_i \pi(i) \log \frac{\pi(i)}{\pi_0(i)} - \sum_i \pi(i) \log \frac{\pi_r(i)}{\pi_0(i)} \\ &= \text{KL}(\pi\|\pi_0) - \frac{1}{\tau} \sum_i \pi(i) r_i + \log Z_r.\end{aligned}$$

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$\pi(i)/\pi_r(i) = [\pi(i)/\pi_0(i)]/[\pi_r(i)/\pi_0(i)]$ . The denominator ratio is divided, so logarithms subtract. Therefore

$$\begin{aligned}\text{KL}(\pi\|\pi_r) &= \sum_i \pi(i) \log \frac{\pi(i)}{\pi_r(i)} \\ &= \sum_i \pi(i) \log \frac{\pi(i)}{\pi_0(i)} - \sum_i \pi(i) \log \frac{\pi_r(i)}{\pi_0(i)} \\ &= \text{KL}(\pi\|\pi_0) - \frac{1}{\tau} \sum_i \pi(i) r_i + \log Z_r.\end{aligned}$$

Multiplying by  $-\tau$  and rearranging yields

$$\sum_i \pi(i) r_i - \tau \text{KL}(\pi\|\pi_0) = \tau \log Z_r - \tau \text{KL}(\pi\|\pi_r).$$

The right side is uniquely maximized at  $\pi = \pi_r$  because KL divergence is nonnegative, with equality only when its arguments agree.

# The Theorem Connects the Social Order to Behavior

Taking ratios in the exponential-tilt formula gives

$$\log \frac{\pi_r(i)}{\pi_r(j)} = \log \frac{\pi_0(i)}{\pi_0(j)} + \frac{r_i - r_j}{\tau}.$$

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- ▶ At positive temperature, fitted reward gaps update the reference policy's log odds. Borda determines the ordering, but not these gaps.
- ▶ As  $\tau \downarrow 0$ , the policy concentrates on reward maximizers—in our baseline model, the Borda maximizers.
- ▶ Thus an example with a unique low-welfare Borda winner becomes an example for aggressively optimized scalar RLHF. We will quantify this distortion later.

# Majority Preprocessing Turns Borda into Copeland

Copeland RLHF replaces each population probability by its majority outcome:

$$H_{ij} := \begin{cases} 1, & P_{ij} > 1/2, \\ 1/2, & P_{ij} = 1/2, \\ 0, & P_{ij} < 1/2. \end{cases} \quad \text{Cop}(i) := \sum_{j \neq i} H_{ij}.$$

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*Technical note.* Exact zero-one targets need not have a finite unconstrained optimum. Replacing  $H_{ij}$  by  $\delta + (1 - 2\delta)H_{ij}$ , where  $0 < \delta < 1/2$ , moves every target into  $(0, 1)$  without changing the Copeland ordering.

# Borda and Copeland Can Select Different Winners

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Copeland counts majority wins:  $a$  has two,  $b$  has one, and  $c$  has none. Majority-preprocessed fitting therefore ranks

$$a \succ b \succ c.$$

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- ▶ A scalar reward model is also a social aggregation rule.
- ▶ In the clean tabular population model, uniform logistic fitting produces the Borda ordering by matching total wins.
- ▶ KL-regularized optimization converts fitted reward gaps into log-odds updates; aggressive optimization therefore selects Borda maximizers.
- ▶ Label preprocessing changes the social objective: fitting probabilities gives Borda, while fitting majority outcomes gives Copeland.

# Thanks!

See you next class!

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